

Topic 6

Trigonometry

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

6.1	Pythagoras theorem
6.2	Trigonometric ratios
6.3	Angles of elevation and bearings
6.4	Sine and cosine rules
6.5	Area of a non-right-angled triangle
6.6	Three-dimensional problems

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Right triangles

Pythagoras gives $\text{hypotenuse}^2 = \text{other side}^2 + \text{other side}^2$. The hypotenuse is opposite the right angle. For a missing shorter side, subtract squared lengths before taking a square root. In 3D, use one right triangle to find a face diagonal, then another for a space diagonal.

Worked example

A 6-8-right triangle has hypotenuse $\sqrt{6^2 + 8^2} = 10$.

Exam check: Check the longest side is the hypotenuse before substituting.

2. Sine, cosine and tangent

For a marked acute angle in a right triangle, $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$. Label sides relative to that angle before choosing. Use inverse trig for an angle and check the calculator is in degrees.

Worked example

Opposite = 5 and adjacent = 12, so $\theta = \tan^{-1}(5/12) = 22.6^\circ$ to one decimal place.

Exam check: Do not confuse an angle of elevation with the angle at the top of a vertical structure.

3. Non-right triangles

The sine rule links side/sine of its opposite angle for an opposite pair. The cosine rule $c^2 = a^2 + b^2 - 2ab \cos C$ uses the included angle C. The area formula is $\frac{1}{2} ab \sin C$ for two sides and the included angle. Sketch and label before selecting a formula.

Worked example

Sides 6 and 8 contain 30° . Area = $0.5 \times 6 \times 8 \times \sin 30^\circ = 12$ square units.

Exam check: With two sides and the included angle, cosine rule is usually the direct route to the third side.

4. Bearings, heights and 3D

Draw a plan view for bearings and a side elevation for height. In a cuboid, the face diagonal may form one side of a right triangle with vertical height. An angle between a line and a plane is measured against the line projection onto that plane.

Worked example

A cuboid has base 3 cm by 4 cm and height 12 cm. Base diagonal = 5 cm; space diagonal = $\sqrt{5^2 + 12^2} = 13$ cm.

Exam check: Show which triangle contains the required angle; a 3D sketch alone can hide the right angle.

5. Exact values and rounding

Recall $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$ and $\tan 45^\circ = 1$, plus the standard 0° , 30° , 45° , 60° , 90° values. Keep full calculator precision through intermediate stages, then round as directed. If a question requests an exact value, retain roots or π where appropriate.

Worked example

In a right triangle with hypotenuse 10 and angle 30° , the opposite side is $10 \sin 30^\circ = 5$.

Exam check: An intermediate rounded angle can noticeably change a later length.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 6 · Trigonometry

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 33 marks

Section A · Multiple choice

Choose one answer for each question.

1. Hypotenuse of sides 5 and 12? [1]

- A 11
- B 12
- C 13
- D 17

2. For opposite 3, hypotenuse 5, $\sin \theta$ is [1]

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $\frac{5}{3}$
- D $\frac{3}{4}$

3. For adjacent 4, opposite 3, $\tan \theta$ is [1]

- A $\frac{4}{3}$
- B $\frac{3}{4}$
- C $\frac{3}{5}$
- D $\frac{4}{5}$

4. Which is the sine rule? [1]

- A $\frac{a}{\sin A} = \frac{b}{\sin B}$
- B $a^2 + b^2 = c^2$ for all triangles
- C $\sin A + \sin B = 1$
- D $a \cos A = b \cos B$

5. Area given a, b and included C is [1]

- A $ab \sin C$
- B $ab/2$
- C $ab \cos C/2$
- D $ab \sin C/2$

6. $\cos 60^\circ$ equals [1]

- A 0
- B $\frac{1}{2}$
- C $\frac{\sqrt{2}}{2}$
- D 1

7. If $\tan \theta = 1$ for an acute angle, θ is [1]

- A 30°
- B 45°
- C 60°
- D 90°

8. Cosine rule with included angle C gives $c^2 =$ [1]

- A $a^2 + b^2 + 2ab \cos C$
- B $a^2 + b^2 - 2ab \cos C$
- C $a^2 - b^2$
- D $ab \cos C$

Section B · Structured questions

9. A right triangle has legs 9 cm and 12 cm.

(a) Calculate the hypotenuse. [2]

(b) Find the sine of the angle opposite the 9 cm side. [2]

(c) Find the angle to one decimal place. [2]

[Total: 6]

10. A vertical tower casts a 20 m shadow. The Sun elevation is 35° .

(a) Write an equation involving the tower height h . [2]

(b) Calculate h to one decimal place. [2]

(c) Explain why tangent is the appropriate ratio. [2]

[Total: 6]

11. Triangle ABC has $AB = 8$ cm, $AC = 10$ cm and angle $BAC = 60^\circ$.

(a) Calculate BC using the cosine rule. [3]

(b) Calculate its area. [2]

(c) State the angle opposite BC. [1]

[Total: 6]

12. A cuboid is 6 cm long, 8 cm wide and 24 cm high.

(a) Find the base diagonal. [2]

(b) Find the space diagonal.

[2]

(c) Find the angle the space diagonal makes with the base, to one decimal place.

[3]

[Total: 7]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. C — 13. Pythagoras gives $\sqrt{5^2+12^2} = 13$.
2. A — $3/5$. Sine relates opposite to hypotenuse.
3. B — $3/4$. Tangent relates opposite to adjacent.
4. A — $a/\sin A = b/\sin B$. Each side is divided by sine of its opposite angle.
5. D — $ab \sin C/2$. Triangle area for two sides and included angle is $ab \sin C/2$.
6. B — $1/2$. The exact cosine value at 60° is $1/2$.
7. B — 45° . The acute angle with equal opposite and adjacent sides is 45° .
8. B — $a^2+b^2-2ab \cos C$. The included angle is subtracted in $c^2 = a^2 + b^2 - 2ab \cos C$.

Section B • Worked solutions

9. a) $\sqrt{9^2+12^2}=15$ cm [2]. b) $9/15 = 3/5$ [2]. c) $\sin^{-1}(3/5) = 36.9^\circ$ [2].

10. a) $\tan 35^\circ = h/20$ [2]. b) $h = 20\tan 35^\circ = 14.0$ m [2]. c) Opposite height and adjacent shadow are known; tangent relates them [2].

11. a) $BC^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 84$; $BC = \sqrt{84} \approx 9.17$ cm [3]. b) $0.5(8)(10)\sin 60^\circ = 20\sqrt{3} \approx 34.6$ cm² [2]. c) Angle BAC [1].

12. a) $\sqrt{6^2 + 8^2} = 10$ cm [2]. b) $\sqrt{10^2 + 24^2} = 26$ cm [2]. c) $\tan \theta = 24/10$, $\theta = 67.4^\circ$ [3].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and examiner report provided by the user. All practice questions are original.